RESEARCH PROPOSAL · UNIVERSIDADE FEDERAL DE GOIÁS

Independent Locating-Dominating Sets on Chordal Graphs

Complexity and structural characterizations

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TODAY

Four parts, then an open discussion

1

The problem

Definition, an example, and the structure every ILD set is forced to have

2

What we know

Related problems in the literature and our results so far

3

Why chordal

A warm-up on split graphs and a map of complexity

4

Open questions

Concrete problems and a proposed plan of attack

1 · THE PROBLEM

Independent locating-dominating sets

A set $S \subseteq V(G)$ is an **ILD set** (Slater & Sewell, 2018) if it is simultaneously:

Independent

no two vertices of S are adjacent

Dominating

every $u \notin S$ has $N(u) \cap S \neq \emptyset$

Locating

$u \neq v \notin S \Rightarrow N(u) \cap S \neq N(v) \cap S$

ILD (decision): does G admit an ILD set at all?

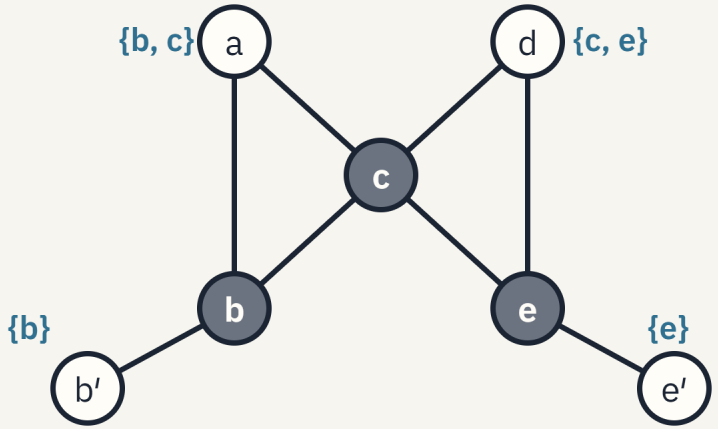
MIN ILD: find one of minimum size, ILD(G).

Unlike LD sets, **existence is not guaranteed**: C_3 and C_4 have none.

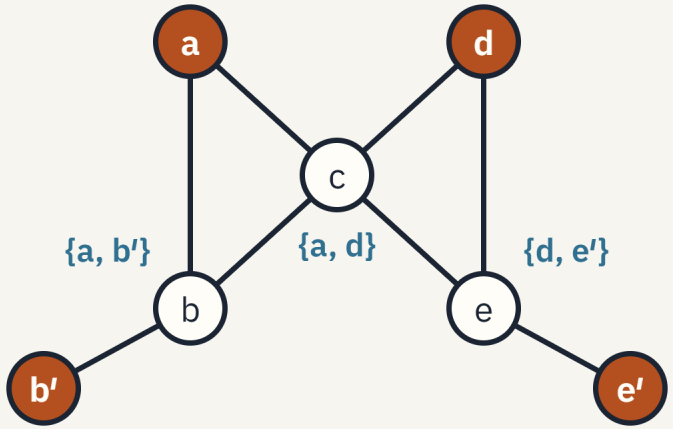
Deciding existence is NP-complete in general.

1 · THE PROBLEM

Independence has a cost



Minimum LD set: {b, c, e}, size 3



The only ILD set: {a, b′, d, e′}, size 4

Every minimum LD set uses the cut vertex c, which no ILD set can contain: with c ∈ S, both a and d would see only {c}.

1 · THE PROBLEM

Independence forces a lot of structure

Cliques

Every clique contains **at most one** vertex of S.

Hence K_n, $n \geq 3$, has no ILD set: the other vertices all see the same {v}.

False twins

If N(u) = N(v), then **both** u and v are in S.

The engine of our NP-hardness gadgets: it pins vertices into S for free.

True twins

If N[u] = N[v], **exactly one** of them is in S.

So a class of three or more true twins rules out any ILD set.

Consequence: the decision question **“does an ILD set exist?”** is already rich, and on chordal graphs, which are built from cliques, these constraints interact strongly.

2

What we know

Neighbouring problems in the literature, and our results on ILD

2 · WHAT WE KNOW

ILD sits between two well-studied problems

Independent domination

- Minimum independent dominating set: linear time on chordal graphs (Farber, 1982)
- Independence + cliques is a friendly combination

ILD

introduced by Slater & Sewell (2018); NP-complete in general

Locating-domination (LD)

- Linear on trees (Slater, 1987) and block graphs (Argiroffo et al., 2020)
- NP-hard on split, bipartite, co-bipartite (Foucaud, 2015)
- NP-complete on interval ∩ permutation graphs (Foucaud et al., 2017)

Key question for chordal graphs: **does ILD behave like independent domination, or like LD?**

Our results so far

POLYNOMIAL

P_4-sparse graphs

Recognition and MIN ILD in $O(n^2)$, via the Jamison–Olariu decomposition.

ETC 2025

NP-COMPLETE

Planar bipartite; planar triangle-free subcubic

Reductions from Planar 3-SAT and P3-SAT_3.

ETC 2026

EXACT VALUES

Graphs built from cycles

Powers C_n^k, $k \geq 2$, admit none; exact values for ladders, Jahangir, helm and sunflower graphs.

Mat. Contemp. 2025

EXACT VALUES

Snarks

Flower snarks: $\lfloor (7n+2)/5 \rfloor$. Generalized Blanuša B^1_n: $3n + 4 - \lfloor (n+2)/4 \rfloor$.

LAWCG 2026

The P_4-sparse and NP-completeness papers were sent in advance; the next two slides only recall the ideas we expect to reuse.

P4-sparse: follow the decomposition

Every P_4-sparse graph is built from K_1 by disjoint union $\oplus$, join $\otimes$ and spider $\triangleright$. Each operation constrains S:

Operation	What happens to an ILD set
$G_1 \oplus G_2$	ILD set exists iff it exists in both; $S = S_1 \cup S_2$
$G_1 \otimes G_2$	S lies on one side and the other side must be K_1; two consecutive joins create true twins, so no ILD set
Spider (R, K, S′)	No clique vertex in S (except the thin spider with $R = \emptyset$); $ILD(G) = \lvert S' \rvert + ILD(G[R])$

Lesson for chordal graphs: the clique side is almost never used; S is pushed onto the independent side.

Hardness: false twins do the work

Planar bipartite

From Planar 3-SAT. Variable gadget H_i, clause gadget Z_j; $|V| = 12n + 4m$.

Planar triangle-free subcubic

From P3-SAT$_3$ (≤ 3 occurrences per variable); $|V| = 8n + 7m$.

Mechanism. False twins x_j, y_j are forced into S; separating them requires the clause vertex to see a literal in S, that is, a true literal.

Obstacle for chordal graphs. These gadgets rely on induced C_4's, and triangle-free chordal graphs are just forests. The reductions do not transfer.

3

Why chordal graphs

Where independence and clique structure meet

Three reasons to look here next

i

A complexity gap to explain

Independent domination is linear on chordal graphs; LD is already NP-hard on split and interval graphs. ILD must land on one side, and the boundary is informative.

ii

Independence meets cliques

Each maximal clique holds at most one vertex of S, and a chordal graph has at most n maximal cliques: strong local constraints, organised by a clique tree.

iii

A rich toolbox and hierarchy

Perfect elimination orderings, clique separators, simplicial vertices; and nested subclasses (trees, block, interval, split, strongly chordal) to calibrate results.

On split graphs, S is almost determined

Let $G = (K, I)$ be split, K a clique and I independent. Independence gives $|S \cap K| \leq 1$.

Case $S \cap K = \emptyset$

A vertex of I outside S has no neighbour in S, so $\mathbf{S = I}$.

Valid iff the sets $N(x) \cap I$, $x \in K$, are non-empty and pairwise distinct.

Case $S \cap K = \{k\}$

Vertices of $I \setminus N(k)$ can only be in S; those in $N(k)$ cannot, and all see $\{k\}$. So $\mathbf{S = \{k\} \cup (I \setminus N(k))}$ and $|N(k) \cap I| \leq 1$.

Then check that the traces of $K \setminus \{k\}$ are distinct.

At most $|K| + 1$ candidates, each checked in polynomial time ⇒ ILD and MIN ILD are polynomial on split graphs, while LD is NP-hard there.

Complexity on the chordal hierarchy

Class	Min. independent domination	Locating-domination	ILD
Trees	linear	linear [Sla87]	always exists [CP26]; min. linear by MSO_1, no explicit algorithm
Block graphs	linear	linear [ABLW20]	linear by MSO_1 metatheorem; explicit algorithm?
Split	linear	NP-hard [Fou15]	polynomial (warm-up, to verify)
Interval	linear	NP-complete [FMNPV17]	open
Strongly chordal	linear	no result found; open even for unit interval [FMNPV17]	open
Chordal	linear [Far82]	NP-hard	open

[CP26] Cáceres & Pelayo (2026): every tree, and every twin-free unicyclic graph, admits an ILD set; MIN ILD on trees left open.

4

Open questions

Where we would like to start working together

4 · OPEN QUESTIONS

Complexity

Q1

Chordal graphs

Is deciding the existence of an ILD set NP-complete? Is MIN ILD harder than existence?

Q2

Interval graphs

LD is NP-complete here. Does independence restore tractability, e.g. by a left-to-right sweep?

Q3

Intermediate classes

Strongly chordal, directed path and k-trees: where exactly is the boundary?

Q4

Explicit algorithms

MIN ILD on trees is stated open by Cáceres & Pelayo (2026). An explicit linear algorithm, then block graphs.

4 · OPEN QUESTIONS

Structural characterizations

Q5

Which chordal graphs admit an ILD set?

A description through the clique tree or forbidden configurations (true-twin triples, “bare” cliques, ...).

Q6

Bounds

Compare ILD(G) with i(G) and LD(G) on chordal graphs; tight upper bounds in terms of n.

Q7

Uniqueness

On split graphs there are at most |K| + 1 ILD sets. When is the ILD set unique? How many can a chordal graph have?

Q8

Simplicial vertices

What role do simplicial vertices play? Can a perfect elimination ordering drive a greedy choice?

4 · OPEN QUESTIONS

Two directions for Q1

If polynomial

- Dynamic programming over a clique tree
- State per bag: which vertex of the clique (if any) is in S
- Difficulty: traces of vertices shared by many bags

If NP-complete

- New gadgets built from chains of cliques
- Split graphs are easy, so hardness needs depth
- Try interval graphs first, adapting the LD reduction

The warm-up suggests one big clique is not enough for hardness; **the interplay between many cliques is where the difficulty must live.**

A possible plan of work

STEP 1

Consolidate

Write up the split-graph result; survey strongly chordal and interval literature

STEP 2

Easy end

Explicit algorithms and characterizations for trees and block graphs

STEP 3

Interval

Settle Q2, in either direction

STEP 4

Chordal

Attack Q1 and Q5 with what steps 2–3 teach us

Let’s discuss

> Which question looks most promising to you?
>
> Do you see a known technique that fits Q1 or Q2?
>
> Can we check the split-graph argument on the board?

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